Test Math Notes

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Chapter 1

Introduction

1.1 Example

Definition 1.1.1. A random variable X is Gaussian if $X \sim \mathcal{N}(\mu, \sigma^2)$.

Theorem 1.1.1. *Let* $X_1, \ldots, X_n \overset{iid}{\sim} \mathcal{N}(\mu, \sigma^2)$. *Then* $\bar{X}$ *is unbiased.*

Proof. By linearity, $\mathbb{E}[\bar{X}] = \mu$. $\square$

1.2 Algorithm Example

Algorithm 1 Gradient Descent

Initialize x_0
for $k = 0, 1, 2, \ldots$ **do**
    $x_{k+1} \leftarrow x_k - \eta \nabla f(x_k)$
end for

1.3 Code Example

```
import numpy as np
import pandas as pd
import scipy as
def gradient_descent(x0, eta, grad, steps):
    x = x0
    for _ in range(steps):
        x = x - eta * grad(x)
    return x
```